

Auteur: Nathan De Roy
Studierichting: Informatica
Oefeningenreeks 4A nr. 42.4

Los volgende integraal op a.d.h.v Fuss: $I = \int \frac{2x+1}{x^2-1} dx$

$$I = \int \frac{2x+1}{x^2-1} dx = \int \frac{2x+1}{(x+1)(x-1)} dx$$

Neem $A, B \in \mathbb{R}$ zodat:

$$I = \int \left(\frac{A}{x-1} + \frac{B}{x+1} \right) dx$$

met:

$$A = \left. \frac{2x+1}{x+1} \right|_{x=1} = \frac{3}{2}$$

$$B = \left. \frac{2x+1}{x-1} \right|_{x=-1} = \frac{1}{2}$$

$$\Rightarrow I = \frac{3}{2} \int \frac{dx}{x-1} + \frac{1}{2} \int \frac{dx}{x+1}$$

$$\Rightarrow I = \frac{3}{2} \ln|x-1| + \frac{1}{2} \ln|x+1| + C$$

References